KO1 AT3.

AI

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1 A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection

a) Intelligent- agents in a smart home:

simple - reflex : uses condition action rules on the current present only.

ex:- If motion detection and its is night, turn on the light. It is fast and cheap but has no memory. So it can't tell whether a room is empty or the user is just sitting still.

Model based reflex :-

keep an internal state of home.

ex : It refers the user is still in the living room Deposite no motion for 5 min

Goal based reflex:.

choose actions to reach a goal such as "room at 24°c by 7pm" or house secured by 11pm. It plans ahead for example planning but does not weight trade-offs.

Utility based.

Maximizes a utility function that balances comfort, energy cost and security. Withes best overall score. It also copes with uncertainty.

Best Choice.

a hybrid with a learning component o reflex rules for immediate safety and security. when expect matter.

- Goal based planning for scheduled routine.
- A utility based layer for trade offs between comfort and energy.

A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. explain now.

Uniform cost Search (UCS).

Expands the node with the lowest path cost (gen) using step costs, It finds the cheapest path. It is complete and optimal if cost are positive.

Interative deepening Search (IDS)

Runs depth - Limited DFS with limit 0, 1, 2, and soon. It is complete, optimal for uniform step costs and uses little memory. It suits an unknown more because the goal depth is not known as pros and cons uses pros and cons.

Handles variable costs. but its memory use is exponential and it explores in the directions.

OIDS pros and cons

Very memory efficient but it re expands upper levels (a small over head since most nodes are at the deepest level and ignores step costs.

Link to agent type.

- A simple reflex agent (eg turn left) will ahead. can loop forever in a maze.
- A model based agent remembers visited cells and builds a map.

A goal based uses search to plan a path to the exist.

A utility based agent chooses among path using time, energy and risk.

Agent design decides what the robot remembers and how it chooses b/w action So the search strategy only works with if the agent has a model of the maze.